

Research Article

Modeling the residence time distribution of integrated continuous bioprocesses

Jure Sencar ¹

Nikolaus Hammerschmidt ¹

Alois Jungbauer ^{1,2}

¹ Austria Centre for Industrial Biotechnology, Vienna, Austria

² Department of Biotechnology, University of Natural Resources and Life Sciences, Vienna;
Muthgasse 18, A-1190 Vienna, Austria

Correspondence: prof. Alois Jungbauer, Department of Biotechnology, University of
Natural Resources and Life Sciences, Vienna; Muthgasse 18, A-1190 Vienna, Austria

E-mail: alois.jungbauer@boku.ac.at

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Abbreviations: **RTD**, residence time distribution; **PDF**, probability distribution function; **FCUO**, fully continuous unit operation; **SCUO**, semi continuous unit operation; **ACC**, alternating column chromatography; **PCC**, periodic counter current chromatography; **CSTR**, continuous stirred tank reactor

Abstract

In response to the biopharmaceutical industry advancing from traditional batch operation to continuous operation, the FDA has published a draft for continuous integrated biomanufacturing [1]. This draft outlines the most important rules for establishing continuous integration. One of these rules is a thorough understanding of mass flows in the process. We developed a computer simulation framework for modeling the residence time distribution (RTD) of integrated continuous downstream processes based on a unit-by-unit modeling approach in which unit operations are simulated one-by-one across the entire processing time, and then combined into an integrated RTD model. The framework allows for easy addition or replacement of new unit operations, as well as quick adjustment of process parameters during evaluation of the RTD model. With this RTD model, the start-up phase to reach steady state can be accelerated, the effects of process disturbances at any stage of the process can be calculated, and virtual tracking of a section of the inlet material throughout the process is possible. A hypothetical biomanufacturing process for an antibody was chosen for showcasing our RTD modeling approach.

1 Introduction

In response to the biopharmaceutical industry advancing from traditional batch operation to continuous operation, the FDA already published a draft for continuous integrated biomanufacturing [1]. This publication outlines the most important rules for establishing continuous integration. One of these rules is a thorough understanding of mass flows in the process, as many unit operations are interconnected. Due to the late advancement of the biopharmaceutical field compared to other fields, recent studies have focused mainly on translating individual unit operations from a batch processing mode to a continuous processing mode [2-5], the development of alternative methods suitable for continuous processing [5-6], and the optimization of the methods [7-17]. Though the individual unit operations are being researched and developed, process integration, especially in terms of process control and process optimization, remains one of the biggest challenges in bringing continuous process operation to the biopharmaceutical industry [17-22].

The analysis of mass flows is already well established in other industries employing continuous manufacturing, such as food industry [23, 24] and the production of small molecules [25, 26]. One of the unique challenges that biopharmaceutical industry presents is the usage of unit operations in which the fluid elements are being processed in a periodical fashion. The aim of this paper is to lay down the theoretical foundations and provide a modeling approach for describing the mass flow in end-to-end integrated continuous downstream processes.

A thorough understanding of mass flow in an end-to-end integrated process operation is needed to estimate the duration of the start-up and shutdown phases, the propagation of possible disturbances, and the fluctuations of process fluid species. A model based on residence time distribution (RTD) can provide such understanding [1]. In addition, an RTD model can be used as a tool for process design and optimization, root cause investigation, batch definition, and scaling up design.

Here, we present a platform for building an RTD model of a fully continuous integrated bioprocess. The platform is based on the volumetric flow of the process fluid and the concentration profiles of its components. The goal of the framework is to produce fast and accurate RTD models based on available data about the process that is to be modeled. Generated RTD models should reflect the understanding of the process in terms of RTD. The RTD model should be easy to update as the additional data and process understanding are continuously harvested during process development. Seamless integration of new knowledge in the RTD model allows the model to be maintained throughout the product lifecycle and plays a role in the continuous improvement of the process.

The RTD model can also be used in linkage studies and process optimization. Such studies can have a large number of free parameters and non-linear behavior. Thus, it is important to integrate as much knowledge as possible into the RTD model in order to narrow down the design space. The RTD model can also serve as a mutual point of understanding among scientists from different fields who work on the same biomanufacturing process. This contributes to more efficient communication and better mutual understanding.

2 Theory

2.1 Residence time distribution

The RTD is a probability distribution function (PDF) of the amount of time that solid or fluid elements spend inside a unit operation or a train of unit operations. The mean residence time is the first moment of the RTD. If an RTD describes the mass flow of a product traveling through a unit operation, then the total area of the RTD corresponds to the yield of the unit operation. A yield <1 indicates that part of the product is lost at that unit operation.

Ideally, the PDF is based on an appropriate experimentally validated mechanistic model that incorporates various design and operation parameters, such as scale, product concentration, process fluid properties, and operating flow rate. However, this is not entirely feasible, as the

mechanistic models can only describe ideal behavior. In addition to the limitations of mechanistic models, limited resources for building and experimentally validating highly sophisticated models push towards alternative approaches. Thus, PDFs based on idealized mechanistic models, or even purely statistical models, are used [27]. Some of the commonly used PDFs are listed in Table 1. The PDFs with more free parameters have greater descriptive power, but also require more experimental data. One option to describe non-ideal unit operations is to use a convolution of two PDFs in order to describe its RTD. In such cases one PDF is describing the idealized behavior of the unit operation, while the other one is describing non-idealities. However, one needs to be careful not to overfit the peak, which could be achieved by introducing constraints between the parameters in order to narrow down the design space.

2.2 Fully continuous unit operation

We define a fully continuous unit operation (FCUO) as a unit operation in which the RTD does not change with time. However, processing time conditions, such as inlet flow rate and process fluid properties, may change, which can impact the RTD. If the impact on the RTD is negligible, then the FCUO can be described by a single PDF. In such cases, passage of the process fluid throughout the FCUO can be simulated by convoluting the inlet concentration profile with the PDF of the FCUO. The result then equals the outlet concentration profile.

Typical FCUOs include tubing, tubular reactors, flow-through columns, continuous stirred tank reactors (CSTRs), inline mixers, continuous dead-end filtration, single or multi-stage tangential flow filtration, and continuous diafiltration. If the outlet flow rate is periodically interrupted, such as when switching between columns, then such unit operation is considered a piecewise FCUO. In this case, the inlet stream is split into intervals by the switch times. The intervals of piecewise FCUO are simulated separately, and then joined into an outlet concentration profile. Maintaining an uninterrupted outlet flow rate profile results in periodic decreases in product concentration due to the broadening of the RTD and an abrupt cutoff due to the switching. Cutting out the concentration decrease results in periodically interrupted outlet flow rate.

2.3 Semi continuous unit operation

We define a semi continuous unit operation (SCUO) as a unit operation in which the RTD changes in a periodical fashion with time. The RTD is directly dependent on the total processing time. For example, the product fraction that enters a chromatographic column at the beginning of a loading phase usually spends more time inside the column than the fraction entering the column towards the end of the loading phase. Thus, the sections experience different RTDs. We define the cycle as the duration of the RTD period; a cycle in multi-column chromatography thus corresponds to the loading time of a single column, which is different from the classical definition where all the columns need to return to their initial position for a complete cycle.

Examples of SCUOs include various implementations of chromatographic capture steps [4,7,19], alternating surge tank systems [28], and centrifugation with periodic discharge of accumulated material [29]. The study of SCUOs is split into two parts: allocation of inlet material among the SCUO cycles, and the description of the outlet peak for individual cycles. In the case of multi-column chromatography, the first part comprises determining which section of the inlet material binds to which column, and the second part would correspond to describing the elution peak. Different types of SCUOs are shown in Figure 1 in the Supporting Information.

2.4 Surge tank

A surge tank accepts an unsteady inlet flow rate and enables a steady outlet flow rate. It is used after a SCUO or a train of synchronized SCUOs. The role of a surge tank is to provide a steady flow rate and homogenize process fluid. A simple variation of the surge tank is a single CSTR. The downside of a single CSTR is its broad RTD. A funnel-like surge tank could be used as an alternative to a CSTR in order to achieve a narrower RTD. However, in both designs, a narrower RTD is achieved at the expense of the capacity to smoothen the fluctuations. These fluctuations in product or modifier concentration stem from the periodic outlet peak collection and could affect the unit operations downstream. Thus, a sufficient homogenization of the process fluid stream might need to be ensured, depending on the subsequent unit operations.

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An alternative approach to a single vessel would be a system of two alternating CSTRs. In such a setup, one CSTR captures and homogenizes the outlet peak from the preceding SCUO as the other CSTR provides a homogeneous feed downstream. Peaks that are discharged into a surge tank are relatively short compared to the cycle time. This provides more flexibility in timing the switching between surge tanks. Such a setup smooths out the fluctuations without broadening the RTD and offers extra time for optional cleaning in place, but requires additional equipment and a more complex operation procedure.

2.5 Process integration

Multiple continuous unit operations are linked together in sequence to form an integrated continuous process. The mass flow at the inlet is determined by the upstream process (e.g., perfusion bioreactor). There are two main approaches to modeling the train of unit operations: the unit-by-unit approach and the time step approach. In the unit-by-unit approach, the propagation of the mass flow is evaluated for the first unit operation across the entire processing time. The outlet mass flow profile of the first unit operation then becomes the inlet mass flow profile for the second unit operation. The resulting profiles for the simulation of the second unit operation then become the inlet profiles for the third unit operation. This is repeated until the end of the process (Figure 2A in the Supporting Information)

The unit-by-unit approach allows unit operations to be simulated independently. Thus, each unit operation can use the optimal evaluation method given its implementation. On the other hand, the feedback control systems spanning back further than one unit operation cannot be implemented directly. In this case, a workaround employing an iterative evaluation algorithm would be needed. In the time step approach, all unit operations are evaluated for a single time step before moving to the next time step (Figure 2B in the Supporting Information). This approach allows for integration of feedback control systems, but also requires each unit operation to be able to process the inlet profile and calculate the outlet profile simultaneously. Moreover, all unit operations need to simultaneously hold information about their current state,

which results in much more complex software implementation and additional computational time. The latter reduces the available temporal resolution of the integrated RTD model.

2.6 Fluctuations and disturbances

SCUOs and piecewise FCUOs typically cause fluctuations in product concentration due to their periodic nature. To design a process in which such fluctuations remain within process specifications, information regarding steady state product concentration variability throughout the entire process is needed. Such information can be estimated as the response of an RTD model to a constant inlet mass flow profile. The initial conditions in the RTD model could be set to the steady state conditions in order to avoid simulating a start-up phase and waiting until the simulating model reaches a steady state. This is especially important when studying such variability at higher temporal resolution (smaller time steps).

Disturbances are caused by unexpected events, which may occur at any stage of the process with various intensity and duration. The disturbances are introduced in the RTD model by modifying the mass flow profile before the selected unit operation. The disturbance then propagates downstream as the simulation progresses throughout the subsequent unit operations. Thus, assessment of the severity and broadening of the disturbance can be performed at any stage of the process.

Delays between the operating cycles among SCUOs affect the fluctuations and the propagation of disturbances. In the case of an integrated process with multiple SCUOs that are not operated in a synchronized fashion, those delays are changing over time. Such changes need to be considered, especially when evaluating a worst-case scenario.

2.7 Chromatographic capture steps

Chromatographic steps operated in semi-continuous mode, such as alternating column chromatography (ACC) and periodic counter current chromatography (PCC), employ the same chromatographic procedures as in batch mode. The integration of chromatographic steps into a

continuous process is enabled by periodic switching of the fluid streams among two or more columns. In the case of twin column ACC, while one column is being loaded, the other undergoes all of the other phases (e.g., wash-elution-regeneration-equilibration). If the loading phase accounts for at least 50% of the cycle duration, then two columns are enough to process a continuous inlet stream. In the case of a shorter loading phase, more columns are required unless the load is gathered before the ACC and loaded in a shorter interval at a higher flow rate. An elution peak is cut based on cut-off criteria and sent towards the next unit operation. If the elution peak is shorter than the loading phase, then the outlet profile consists of periodic peaks encapsulated in a periodic box-shaped flow rate profile.

The amount of material that binds to the column is determined by the breakthrough profile analogous to traditional batch chromatography. However, the breakthrough profile does not provide detailed information about the interaction between the ligands and the products and would not distinguish between a case of strong binding and a case of strong displacement. In the first case, the breakthrough product originates from the late part of the load, and in the second case the breakthrough product originates from the early part of the load (Figure 1). Though the breakthrough profile remains the same in both cases, following a section of the inlet material would result in different RTDs.

The elution peak shape is predicted as in batch mode. If the elution peak shape stems from a computationally demanding simulation, the computation time of the model can be optimized by pre-calculating peak shapes. The pre-calculated peak shapes are repeated for each cycle and adjusted by the parameters influencing the peak shape, particularly the amount of gathered material within each cycle. The elution peak is then cut based on cut-off criteria and fed into the next unit operation. Thus, the outlet stream is periodically turned on and off. In cases where the distribution of the material within the peak broadly affects the RTD downstream, a more complex model describing the distribution of the captured material within the elution peak may need to be established.

2.8 Tracking a section of product

Following a section of the inlet material is crucial for simulating disturbances and estimating an end-to-end processing time in a steady-state regimen. In order to track a section of product, it needs to be labeled. The labeled section is simulated as a separate component with the same properties as the rest of the product. The unit operations, for which the RTD depends on the product concentration, need to be able to process multiple components simultaneously. A list of components with the same properties needs to be specified for such unit operations. The unit operation, for which the RTD does not depend on the product concentration, can process labeled and unlabeled components separately.

2.9 Accelerated start-up phase

The start-up phase of a process can be assessed as a response of the RTD model to an input step function. The step function can be replaced by a known start-up profile of the upstream process (e.g., perfusion bioreactor) or any other preceding unit operations. The process may employ an accelerated start-up phase by having some unit operations, such as surge tanks empty or only partially pre-filled at the beginning of the operation. Such a surge tank has a sharper onset in the outlet product concentration profile and exhibits a shorter start-up phase. For some unit operations, a narrower onset could be achieved by pre-flushing the unit with a buffer with different properties (e.g., lower density) than the process buffer. In such a case, the accelerated start-up phase should be simulated separately from the steady state operation, as it is governed by different dynamics.

The start-up phase could be further optimized by specifying the conditions under which the unit operation is connected to the inlet stream. For example, reaching a certain product concentration before the inlet stream is redirected into a surge tank results in a higher initial concentration, which can in turn lead to a faster start-up phase. Analogously, the conditions under which the outlet stream gets released towards the next unit can be defined.

2.10 Computing time

A fast evaluation of the RTD model allows for various optimization techniques and is needed to build a responsive graphical user interface that allows the user to adjust the values of select process parameters. In order to speed up the evaluation, the PDFs derived from the computationally demanding simulations can be pre-calculated. An example of such a PDF is a chromatographic peak shape obtained by the computationally demanding numerical simulation of a mechanistic model. To describe the dependency of pre-calculated PDFs on process parameters, the PDFs can be pre-calculated under different conditions and described by a statistical or engineering function. For cases in which it is too challenging to find an appropriate descriptive function, interpolation can be used instead.

2.11 The level of detail in the RTD model

The task of the modeling platform is to offer or allow the inclusion of new RTD models given the available data. Thus, the level of detail in the RTD model depends primarily on the data and process understanding. For example, a chromatographic peak shape may be derived from a mechanistic model or, more simply, described by fitting an exponentially modified Gaussian curve to the experimental data. The RTD modeling platform should be capable of describing both variants and integrating them into the appropriate RTD model. In other words, a good RTD model should have sufficient flexibility to describe the process in a way that matches our overall understanding of that process.

3 Methods

3.1 Software

The code was written in the Python 3.7+ programming language, and the Bokeh visualization library was used for plotting and creating a graphical user interface.

3.2 Modeling approach

We decided to use the unit-by-unit modeling approach, in which unit operations are simulated one-by-one across the entire processing time. This allows for a more intuitive workflow, easy swapping and modification of unit operations, faster calculation, and higher temporal resolution.

3.3 Assumptions

In order to balance the added value, the required data, and the complexity of the modeling approach, we included the following assumptions in our RTD modeling approach: 1) FCUOs have a constant RTD; 2) all unit operations except for surge tanks have a constant inlet flow rate with optional initial delay due to the start-up dynamics of the preceding unit operation; 3) SCUOs produce a periodic box-shaped outlet flow rate profile at steady-state operation; 4) SCUOs produce homogeneous peaks; and 5) the RTD of FCUOs with a void volume $<1\%$ of the void volume of surrounding unit operations is neglected. Assumptions 1 and 2 allow for the use of convolution, which simplifies the description of the unit operations and improves the computation time. Assumption 3 allows surge tanks following the SCUOs, providing a constant outlet flow rate profile for the subsequent unit operations. The implication of assumption 4 is that the outlet peak of the SCUOs depends only on the amount of collected material within the cycle and not on the order in which the inlet material enters the cycle. The assumption does not have a significant impact on the RTD if at least one of the following conditions is met: the duration of the peak recovery is much shorter than the cycle duration, the material is homogenized within the SCUO (via mixing, dispersion, or diffusion), there is separation between

product isoforms¹, or the material is homogenized after the SCUO (e.g., by alternating surge tank systems).

A case in which assumption 4 could potentially lead to a significant impact on the RTD would be a chromatographic step with weak binding and long elution time, followed by a small single surge tank. In such a case, simulation of the displacement, diffusion, and dispersion of the product may be needed. Furthermore, the part of the model describing the dynamics would need to be experimentally validated by labeling select sections of the inlet material and analyzing the elution peak within the design space.

Assumption 5 allows unit operations to be excluded, such as small dead-end filters and small tubing from the RTD model, as they do not have a significant impact on the overall RTD. In the case of dilution via a small inline mixer, the mass flows are simply added together.

In the case of chromatographic steps, we used the constant pattern solution [30] to describe a load breakthrough profile. In addition, we used the following assumptions: binding capacity is independent of inlet concentration, and the displacement effect is negligible. The first assumption allows a variable inlet concentration profile. We expect the fluctuations in the steady state inlet concentration profile to be sufficiently small, so they do not affect the binding capacity. Very low concentrations are expected only at the beginning of the start-up phase. The second assumption allows us to determine which parts of the inlet profile are captured on the column. Furthermore, the RTD of the uncaptured product traveling through the column can be approximated by the RTD of the product traveling through the column under non-binding conditions. We can measure the latter in a pulse injection experiment. With this information, we can track a labeled section of material throughout the chromatographic steps. Once again, in the

¹ Separation between product isoforms redistributes the product based on isoforms rather than the time at which fractions of the product enter the unit operation. An example would be a chromatographic step employing linear gradient elution for polishing product isoforms.

case of different dynamics, appropriate extension to the RTD model and additional experiments to characterize such dynamics would be needed.

The framework is designed to be expanded in the case that any of the limitations need to be overcome.

3.4 Fully continuous unit operation

FCUOs have a constant RTD and constant inlet and outlet flow rates. Thus, the mass flow is simulated by convoluting the inlet concentration profile with the RTD of the FCUO (Figure 2A). The convolution assumes that the initial concentration values in the FCUO are all zero. Custom initial conditions are required in cases in which the initial concentrations are not zero or for simulating a steady-state regimen. Custom initial conditions are set by prepending a section with the initial concentrations in the inlet concentration profile. The length of the prepended section should be the same as the length of the RTD of the FCUO (Figure 2B). In other words, we simulate pre-flushing the unit operation.

In the case of piecewise FCUO, the inlet profile is split into sections. Each section belongs to one cycle and is optionally prepended and appended by a specified concentration profile (e.g., the one from the equilibration buffer) and convoluted with the RTD. The resulting profile for each section is trimmed at the outlet switch times and joined together into the resulting outlet profile (Figure 2C).

3.5 Chromatographic capture steps

In batch chromatography, the load is typically homogenized, which ensures a constant product concentration in the load. However, in continuous operation, the feed can have a variable product concentration. Thus, the traditional assessment of the amount of material captured on the column needs to be extended in order to account for the variability in the product concentration in the feed.

In order to determine which parts of the feed bind to the column, the x-axis of the product concentration in the feed and the breakthrough profile are translated from time scale to the cumulative amount of the material in the feed (Figure 3 in the Supporting Information). The two profiles are then multiplied, and the resulting profile corresponds to the parts of the product bound to the resin. The same procedure can be used for process fluid with multiple components that have the same binding properties. The bound part of the product then proceeds to the wash-elution phase, in which the elution peak is simulated as in batch mode.

To determine the mass flow in the PCC, the model for the ACC is extended by simulating propagation of the unbound material through the column and prepending the resulting profile to the inlet section of the following cycle (Figure 3). In the case of strong binding, the propagation of unbound material is simulated by convoluting the profile of the unbound inlet material with the RTD of the product traveling through the column under non-binding conditions. The resulting profile is cut at the start and end of the cycle time interval, as this is the interval at which the outlet of the first column is connected to the outlet of the second column. Thus, the trimmed section is prepended to the inlet profile of the next cycle. The product before the cut represents the losses due to the product leaching from the second inline column. The product after the cut represents the losses due to column switching.

3.6 Ideal continuous stirred tank reactor

The ideal CSTR is a surge tank with ideal mixing, which results in the concentration at the outlet matching the concentration inside the CSTR. In fully continuous mode, the inlet and outlet flow rates are the same, and the fill level remains constant. In such a case, its RTD can be described by an exponential drop function and the convolution can be used for its evaluation.

In the case of an unsteady inlet flow rate, the volume in the CSTR changes. Due to the changes in fill volume, the RTD changes with time and convolution can no longer be used to simulate propagation of the process fluid. Instead, the set of differential equations describing the contents of the CSTR needs to be solved (Eq. 1-3).

$$\dot{V}(t) = \phi_{in}(t) - \phi_{out}(t) \quad \text{Eq. 1}$$

$$\dot{m}(t) = \phi_{in}(t) * c_{in}(t) - m(t)/V(t) * \phi_{out}(t) \quad \text{Eq. 2}$$

$$c(t) = m(t)/V(t) \quad \text{Eq. 3}$$

Where t is the processing time, $V(t)$ is the process fluid volume in the CSTR, $c_{in}(t)$ is the concentration of observed species in the inlet stream, $m(t)$ is the amount of observed species inside the CSTR, and $\phi_{in}(t)$ and $\phi_{out}(t)$ are inlet and outlet flow rates, respectively. The concentration inside the CSTR and the outlet concentration are the same and denoted by $c(t)$.

3.7 Surge tank systems

The role of a surge tank system is to ensure a steady flow and homogenize the process fluid. In the case of an ideal CSTR, the only process parameter we can vary is the fluid volume inside the CSTR. A CSTR with larger volume provides better homogenization at the expense of broader RTD (Figure 4A-D). Depending on the use-case, a non-ideal CSTR could be approximated as an ideal CSTR with a reduced volume.

An ideal RTD describing efficient homogenization of the outlet peak while keeping cycles separated would have a box shape. Such an RTD could be achieved by a system of two or more alternating ideal CSTRs. In such a system, one CSTR collects and homogenizes the incoming peak while the other provides a steady stream for the subsequent unit operation. When the latter is drained, the order of the two reactors is switched (Figure 4 E,G). A slight overlap between the boxes is expected due to a non-ideal switching mechanism and band broadening in the connecting tubes. The difference between the cycle duration and the peak collection time within the cycle offers a window within which we must switch from one surge tank to another. This window allows complete depletion of each surge tank, though there may be some variability in volumes collected between the cycles. The window also allows for fine tuning the concentration by adjusting the dilution of the collected peak in order to ensure a steady concentration profile downstream (Figure 4F, H).

4 Results

As a showcase, we created an RTD model of an integrated process consisting of constant inlet profile, depth filtration, PCC, surge tank in form of a single CSTR, inline virus inactivation, flow-through chromatography, and ultrafiltration–diafiltration (UFDF). We simulated a process fluid flow and product concentration profile throughout the train of unit operations (Figure 5A). A section of the inlet material was labeled in order to visualize how it propagated through the system. The labeled section can mimic a disturbance that occurs at the inlet. The disturbance could also be introduced at later stages of the process by modifying the corresponding flow rate and concentration profiles.

In this example, we simulated inline periodic switching of depth filters, anion exchange chromatographic (AEX) columns, and UFDF modules. Thus, the unit operations were rendered as piecewise FCUOs. A potential challenge in process development arising from such unit operations is the overlapping of negative spikes in the product concentration.

The surge tank in the example is set to a 10% safety margin from being fully depleted. The actual volume of the surge tank is evaluated at runtime based on the inlet flow rate profile. The surge tank contributes most to the broadening of the RTD of the labeled section of the material, which can represent a potential disturbance stemming from upstream processing.

Variations of the process were evaluated in terms of the width of the disturbance, the duration of the start-up phase and the mean residence time of the disturbance. We define the width of the disturbance as a time interval during which the concentration of the labeled material surpasses 10 % of its maximum concentration. We define start-up time as a time needed for the concentration at the outlet of the process to reach 95 % of the steady state concentration. These definitions were chosen specifically for the given example. Variations of the process include replacing single CSTR with a system of 2 alternating CSTRs (Figure 5), doubling the size of the PCC columns (Figure 6) and pre-filling the CSTR with process buffer without the product (Figure 4 in the Supporting Information). In the process with two alternating CSTRs, one of them

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captures the effluent from the preceding unit operation while the other feeds the following unit operation; once the latter is empty they switch function. . The switch times have a delay of 20 % of the cycle duration relative to the beginning of the peak collection at the preceding unit operation.

Base process (single CSTR and small PPC columns) has a start-up time of 2 h 50 min, the mean residence time of 2 h 10 min and the disturbance width of 2 h 50 min. The variation with 2 alternating CSTR (Figure 5 B) has a longer start-up phase (4 h 40 min) and a longer mean residence time (3 h 50 min) due to larger hold up volume, but a potential disturbance is 40 % narrower (2 h) due to complete depletion of a CSTR within each cycle.

The process with larger PCC columns (Figure 6 B) has longer PCC cycles and in turn higher variability of the product concentration. The start-up time (4 h 20 min), the mean residence time (2h 40 min) and the disturbance width (4 h 30 min) are prolonged. The disturbance has a lower amplitude due to the broadening.

Process with pre-filled CSTR (Figure 4 in the Supporting Information) has a longer start-up phase (4 h 25 min) but the same residence time and the disturbance width as the base process. This is expected as the pre-fill only influences the start-up phase.

5 Discussion

In early stages of development, a less sophisticated RTD model is sufficient to aid in process development. However, adding additional unplanned features to the existing model can quickly lead to tangled code and non-intuitive user experience, even if the code is well documented. Therefore, before building an RTD model, it is important to determine which process scenarios are to be simulated, which features are required, and which features may be needed in the future. On the other hand, it is also important to design the platform in a way that allows for easy upgrades and modifications. Planning ahead is crucial to optimizing the selection and extent of experiments needed for model validation.

Though we only presented a few variations of the RTD model, it can serve as a basis for many more analyses, such as the impact of different delays among unit operations, different start-up conditions, flow rate variations, and different column sizes. Such parameters can also be optimized within the design space. Such optimization can be performed in-silico using a fitness function which can depend on various factors, such as yield, start-up time, critical process parameters, safety margins, and processing time.

Simulation time for presented models was around 80 ms (with a simulation time step of 0.5 s) using a single core on Intel(R) Core(TM) i7-7700 CPU @ 3.60GHz processor. Because of the fast evaluation time, the RTD model can be used for Monte Carlo simulation studies of the variability of process parameters in the RTD. Some process parameters, such as column binding capacity and column packing performance, have inherent variability, and the implication needs to be assessed at later stages of process development. On the other hand, the code also needs to be reliable. Assertions and log messages might slightly impact the performance, whereas external tests that validate each section of the code provide additional reliability without impacting the performance. Such tests are also generally required by the FDA during software validation [31].

In conclusion, our RTD model provides an in-depth understanding of the RTD of an integrated continuous biomanufacturing process as suggested by the draft guidelines of the FDA [1]. In addition, the RTD model can be used in process development and process optimization. The effects of an accelerated start-up phase, the propagation of disturbances, fluctuation of process buffer components, buffer consumption, and equipment size can be simulated. The RTD model generated by the platform can be expanded to include additional critical process attributes derived from models of individual unit operations. A fast evaluation allows for the use of numerical optimization techniques and a quick response of the model upon adjusting process parameters during visualization. The proposed platform is designed to be extended for specific cases and maintained throughout the process life cycle. Our code, which includes RTD modeling

library, testing library for quality assurance of the code in the RTD library and examples is publicly available under MIT open source license [32].

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Conflict of interest

The authors declare no conflict of interest.

7 References

- [1] Quality Considerations for Continuous Manufacturing, Guidance for Industry (2019), Draft Guidance, FDA (15276548 Dft, Feb 2019)
- [2] Martins, D.L., Sencar, J., Hammerschmidt, N., Tille, B., Kinderman, J., Kreil, T.R., and Jungbauer, A. (2019) Continuous Solvent/Detergent Virus Inactivation Using a Packed-Bed Reactor. *Biotechnology Journal*, 14 (8), 1800646.
- [3] Rosa, P.A.J., Azevedo, A.M., Sommerfeld, S., Bäcker, W., and Aires-Barros, M.R. (2012) Continuous aqueous two-phase extraction of human antibodies using a packed column. *Journal of Chromatography B*, 880, 148–156.
- [4] Steinebach, F., Müller-Späth, T., and Morbidelli, M. (2016) Continuous counter-current chromatography for capture and polishing steps in biopharmaceutical production. *Biotechnology Journal*, 11 (9), 1126–1141.
- [5] Burgstaller, D., Krepper, W., Haas, J., Maszelin, M., Mohoric, J., Pajnic, K., Jungbauer, A., and Satzer, P. (2018) Continuous cell flocculation for recombinant antibody harvesting. *Journal of Chemical Technology & Biotechnology*, 93 (7), 1881–1890.
- [6] Hammerschmidt, N., Hintersteiner, B., Lingg, N., and Jungbauer, A. (2015) Continuous precipitation of IgG from CHO cell culture supernatant in a tubular reactor. *Biotechnology Journal*, 10 (8), 1196–1205.
- [7] Baur, D., Angarita, M., Müller-Späth, T., and Morbidelli, M. (2016) Optimal model-based design of the twin-column CaptureSMB process improves capacity utilization and productivity in protein A affinity capture. *Biotechnology Journal*, 11 (1), 135–145.
- [8] Burgstaller, D., Jungbauer, A., and Satzer, P. (2019) Continuous integrated antibody precipitation with two-stage tangential flow microfiltration enables constant mass flow. *Biotechnol. Bioeng.*, 116 (5), 1053–1065.

- [9] Gjoka, X., Gantier, R., and Schofield, M. (2017) Transfer of a three step mAb chromatography process from batch to continuous: Optimizing productivity to minimize consumable requirements. *Journal of Biotechnology*, 242, 11–18.
- [10] Godawat, R., Konstantinov, K., Rohani, M., and Warikoo, V. (2015) End-to-end integrated fully continuous production of recombinant monoclonal antibodies. *Journal of Biotechnology*, 213, 13–19.
- [11] Ichihara, T., ITO, T., Kurisu, Y., Galipeau, K., and Gillespie, C. (2018) Integrated flow-through purification for therapeutic monoclonal antibodies processing. *MAbs*, 10 (2), 325–334.
- [12] Jungbauer, A. (2019) Continuous Virus Inactivation: How to Generate a Plug Flow. *Biotechnology Journal*, 14 (2), 1800278.
- [13] Li, Z., Gu, Q., Coffman, J.L., Przybycien, T., and Zydney, A.L. Continuous precipitation for monoclonal antibody capture using countercurrent washing by microfiltration. *Biotechnology Progress*, n/a (n/a), e2886.
- [14] Rosa, P.A.J., Azevedo, A.M., Sommerfeld, S., Mutter, M., Bäcker, W., and Aires-Barros, M.R. (2013) Continuous purification of antibodies from cell culture supernatant with aqueous two-phase systems: From concept to process. *Biotechnology Journal*, 8 (3), 352–362.
- [15] Senčar, J., Hammerschmidt, N., Martins, D.L., and Jungbauer, A. (2020) A narrow residence time incubation reactor for continuous virus inactivation based on packed beds. *New Biotechnology*, 55, 98–107.
- [16] Vogg, S., Ulmer, N., Souquet, J., Broly, H., and Morbidelli, M. (2019) Experimental Evaluation of the Impact of Intrinsic Process Parameters on the Performance of a Continuous Chromatographic Polishing Unit (MCSGP). *Biotechnology Journal*, 14 (7), 1800732.

[17] Walch, N., and Jungbauer, A. (2017) Continuous desalting of refolded protein solution improves capturing in ion exchange chromatography: A seamless process. *Biotechnology Journal*, 12 (6), 1700082.

[18] Hong, M.S., Severson, K.A., Jiang, M., Lu, A.E., Love, J.C., and Braatz, R.D. (2018) Challenges and opportunities in biopharmaceutical manufacturing control. *Computers & Chemical Engineering*, 110, 106–114.

[19] Jungbauer, A., and Hammerschmidt, N. (2017) Integrated Continuous Manufacturing of Biopharmaceuticals, in *Continuous Manufacturing of Pharmaceuticals*, Wiley-Blackwell, pp. 247–281.

[20] Karst, D.J., Steinebach, F., and Morbidelli, M. (2018) Continuous integrated manufacturing of therapeutic proteins. *Current Opinion in Biotechnology*, 53, 76–84.

[21] Somasundaram, B., Pleitt, K., Shave, E., Baker, K., and Lua, L.H.L. (2018) Progression of continuous downstream processing of monoclonal antibodies: Current trends and challenges. *Biotechnol. Bioeng.*, 115 (12), 2893–2907.

[22] Steinebach, F., Ulmer, N., Wolf, M., Decker, L., Schneider, V., Wälchli, R., Karst, D., Souquet, J., and Morbidelli, M. Design and operation of a continuous integrated monoclonal antibody production process. *Biotechnology Progress*, 33 (5), 1303–1313.

[23] Cavero, C., Dias, E., Gut, J. (2010) Residence time distribution in holding tubes using generalized convection model and numerical convolution for non-ideal tracer detection. *Journal of Food Engineering*, 98, 248–256.

[24] Torres, A.P., Oliveira, F.A.R. (1998) Residence time distribution studies in continuous thermal processing of liquid foods: a review, *Journal of Food Engineering*, 36 (1), 1–30.

[25] Bostijn N., Van Renterghem, J., Vanbillemont, B., Dhondt, W., Vervaet, C., De Beer, T., (2019) Continuous manufacturing of a pharmaceutical cream: Investigating continuous powder

dispersing and residence time distribution (RTD), *European Journal of Pharmaceutical Sciences*, 132, 106-117.

[26] Yijie, G., Fernando, J.M., Marianthi G.I. (2012) A review of the Residence Time Distribution (RTD) applications in solid unit operations, *Powder Technology*, 228, 416-423.

[27] Coker, A.K., and Kayode, C.A. (eds.) (2001) Chapter Eight - Residence Time Distributions in Flow Reactors, in *Modeling of Chemical Kinetics and Reactor Design*,

[28] Schofield, M., and Johnson, D. (2018) Continuous Low-pH Virus Inactivation: Challenges and Practical Solutions. *Genetic Engineering & Biotechnology News*, 38 (15), S13–S15. Gulf Professional Publishing, Woburn, pp. 663–761.1.

[29] Tebbe, H., Lütkemeyer, D., Gudermann, F., Heidemann, R., and Lehmann, J. (1996) Lysis-free separation of hybridoma cells by continuous disc stack centrifugation. *Cytotechnology*, 22 (1–3), 119–127.

[30] Carta, G., and Jungbauer, A. (2010) *Protein chromatography: process development and scale-up*, Wiley-VCH, Weinheim.

[31] *General Principles of Software Validation; Final Guidance for Industry and FDA Staff* (2002), FDA

[32] Sencar, J., (2020) GitHub Repository, <https://github.com/open-biotech/bio-rtd>.

Table 1. Commonly used probability distribution functions.

Exponential decay	$E(t) = \frac{1}{\tau} e^{-t/\tau}$	τ - mean residence time
Laminar flow in tube	$E(t) = \begin{cases} \frac{2t^2}{\tau^3} & t \geq \frac{\tau}{2} \\ 0 & t < \frac{\tau}{2} \end{cases}$	τ - mean residence time
Tanks in series	$E(t) = \frac{t^{n-1} e^{-t/\tau}}{(n-1)! \tau^n}$	τ - mean residence time of a single tank n - number of tanks
Gaussian distribution	$E(t) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{t-\tau}{\sigma}\right)^2}$	τ - mean residence time σ - standard deviation
Exponentially modified Gaussian distribution	$E(t) = \frac{\lambda}{2} e^{\frac{\lambda}{2}(\lambda\sigma^2 - 2(t-\mu))} \operatorname{erfc}\left(\frac{\lambda\sigma^2 - (t-\mu)}{\sigma\sqrt{2}}\right)$ $\mu = \tau - 1/\lambda$	τ - mean residence time σ - standard deviation λ - exponential skew
Skewed normal distribution	$E(t) = \frac{1}{\omega\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{t-\mu}{\omega}\right)^2} * \left(1 + \operatorname{erf}\left(\frac{\lambda(t-\mu)}{\omega\sqrt{2}}\right)\right)$ $\mu = \tau - \omega\lambda\sqrt{2/\pi(1+\lambda^2)}$	τ - mean residence time ω - broadness λ - skew

Figure legends

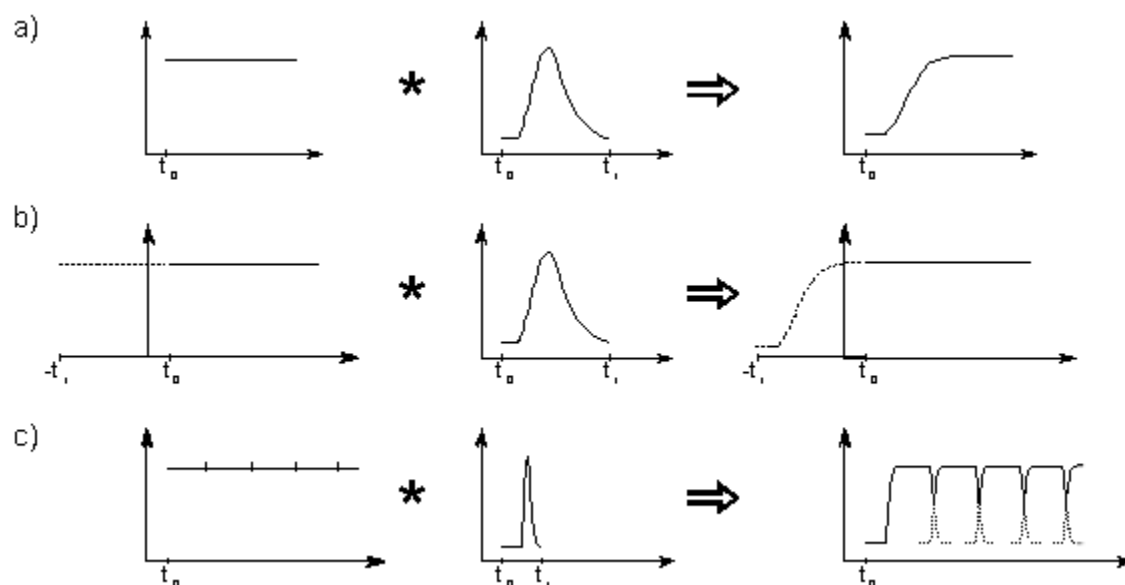


Figure 1. Simulation of the mass flow through a fully continuous unit operation (FCUO). **(A)** The FCUO has a constant residence time distribution (RTD) profile (middle plot) and can be convoluted with the inlet product concentration profile (left plot) to obtain the outlet profile (right plot). **(B)** In order to simulate the initial conditions, we need to prepend a section to the inlet profile. **(C)** A piecewise FCUO has steady operation between cycles; thus, the outlet profile is the sum of all inlet sections convoluted with the RTD of the FCUO. The area under the dotted line represents the product losses due to the combination of band broadening and abrupt switching.

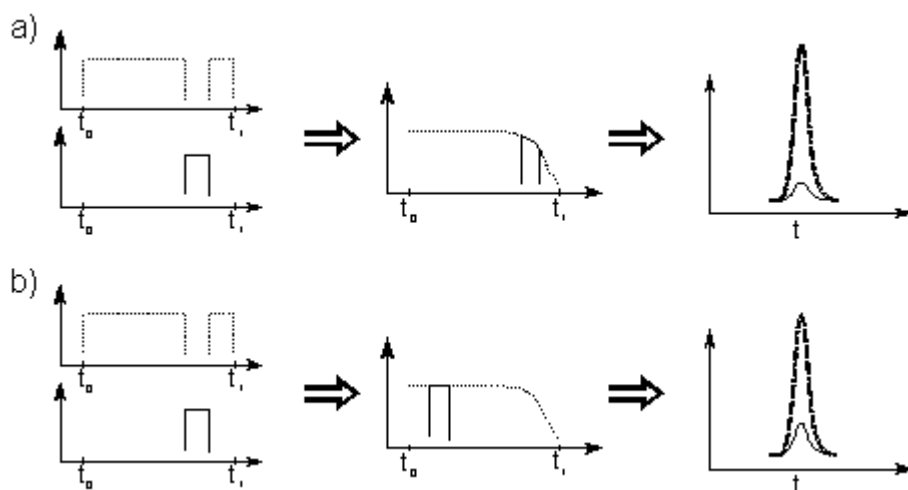


Figure 2. Tracing a labeled product in the chromatographic cycle. Labeled product is simulated as a separate species with the same properties as the unlabeled material. The profiles are summed and evaluated against the loading breakthrough profile in order to determine the amount of bound product inside the column. The elution peak is split between species based on the captured amount. **(A)** In the case of strong binding, the section of material loaded towards the end of the loading phase is breaking through. **(B)** The same section remains entirely within the column in the case of predominant displacement.

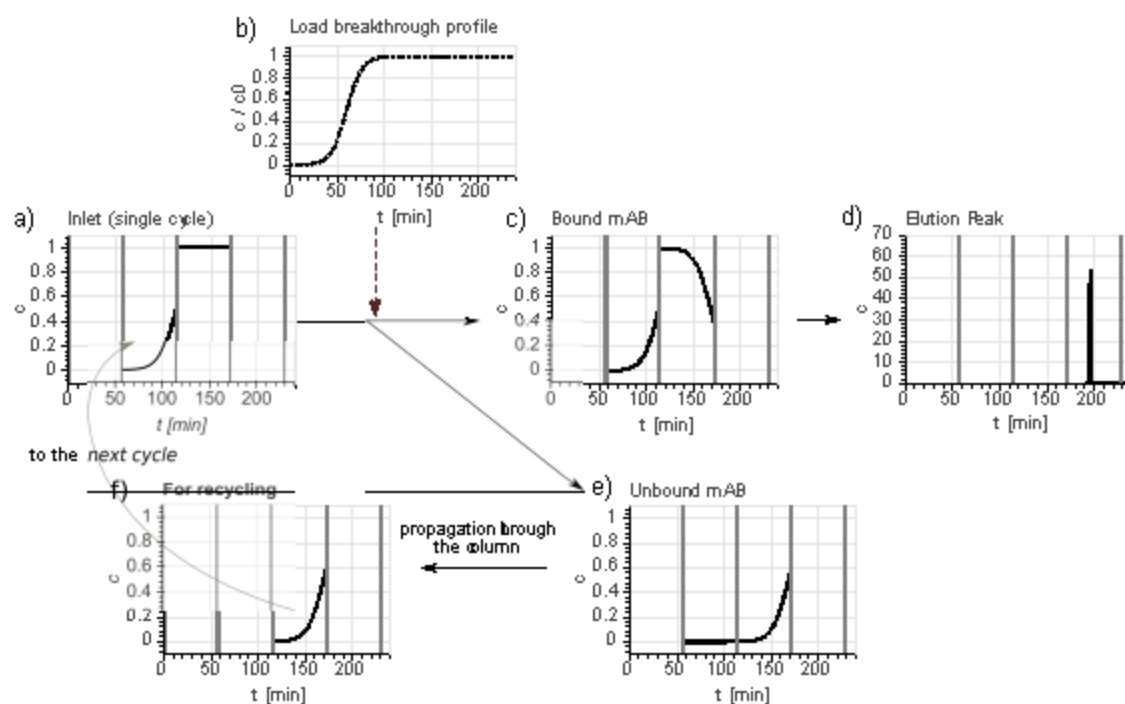


Figure 3. Mass flow of the product through the PCC. The inlet material **(A)** is loaded onto the column. Based on the breakthrough profile **(B)** and the binding properties, the load profile splits between bound **(C)** and unbound profiles **(D)**. The bound (captured) part undergoes the wash-elution step which results in elution peak **(E)**, whereas the unbound part propagates through the column **(F)** and is partially captured by the next column.

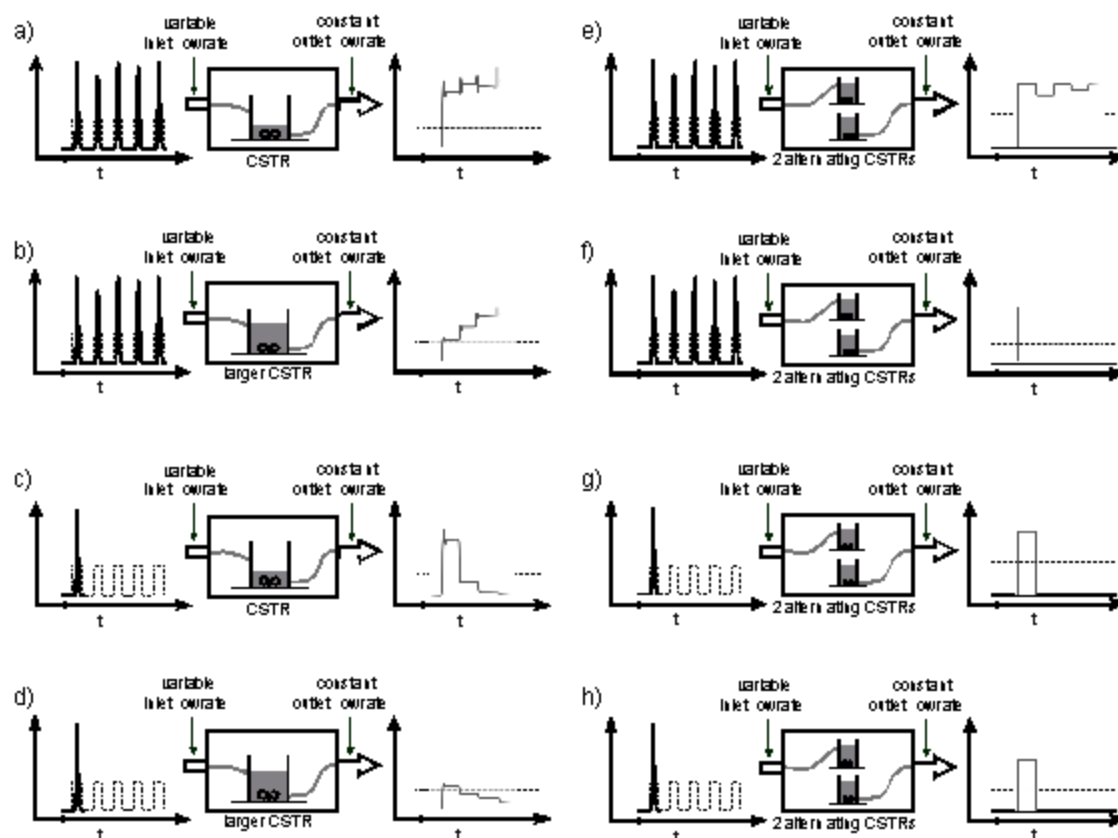


Figure 4. Surge tank systems. **(A, B)** Using a single CSTR as a surge tank results in spikes due to fluctuations in the concentration in collected peaks. **(C, D)** The profile of the observed section of the product has exponential tailing at the outlet of the CSTR. A larger CSTR provides a more stable concentration profile at the expense of longer exponential tailing. **(E)** A system of two alternating CSTRs provides a constant outlet profile with no fluctuations due to peak collection and **(G)** no exponential tailing. **(F, H)** In the case of inconsistent peaks, the concentration within the CSTR collecting the peak can be adjusted by adaptive dilution.

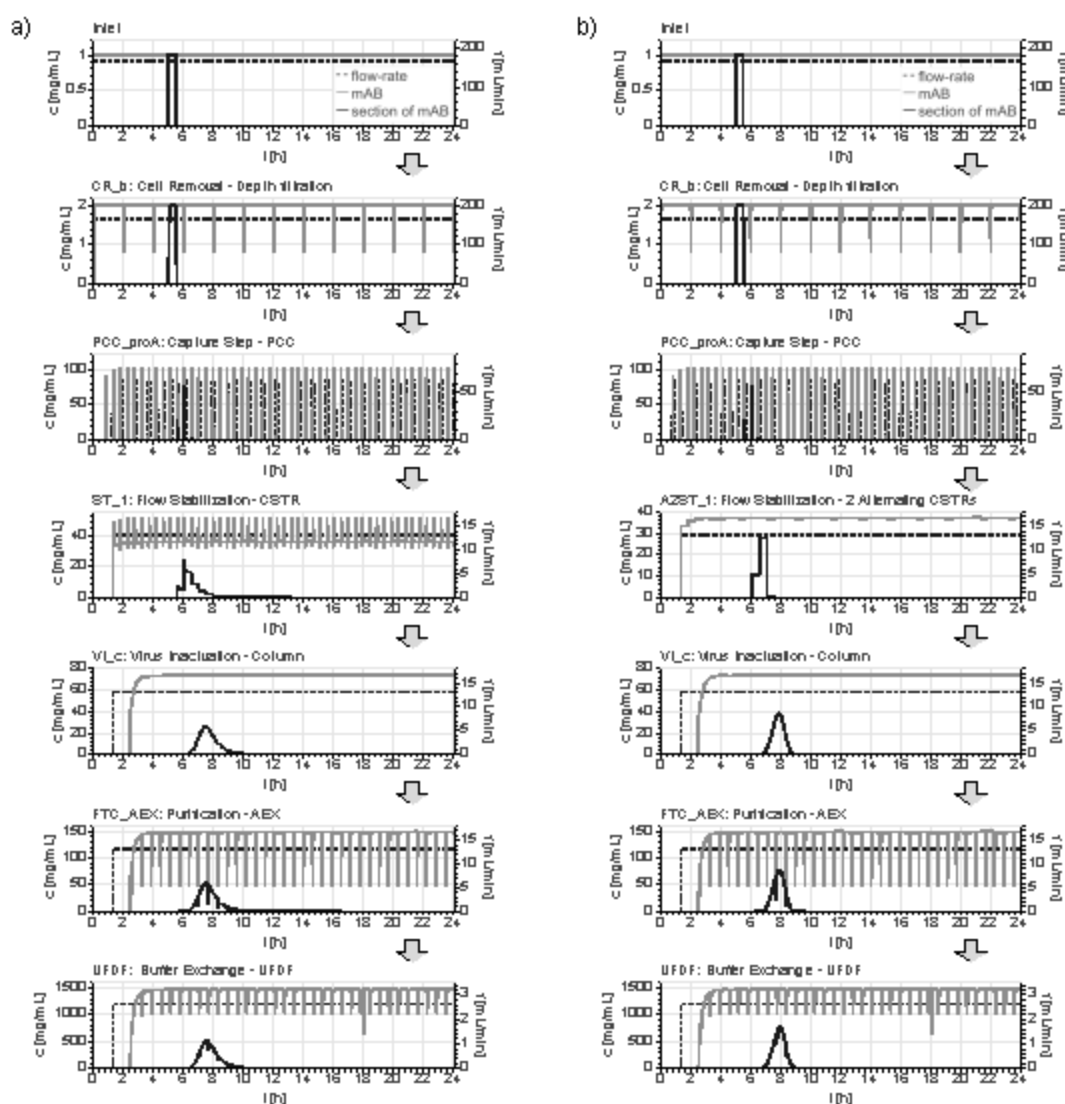


Figure 5. Simulated RTD of an integrated end-to-end continuous downstream process using a single CSTR **(A)** or a system with two alternating CSTRs **(B)**. The upstream feed is set to constant product concentration (solid gray line) and constant flow rate (dashed line). A section of the inlet material (solid black line) is labeled in order to demonstrate the spread of the potential disturbance throughout the process. A train of unit operations is connected to the feed. The propagation of the material from the feed is simulated throughout the process. The duration of the start-up phase, the product concretion fluctuations can be assessed from the resulting profiles.

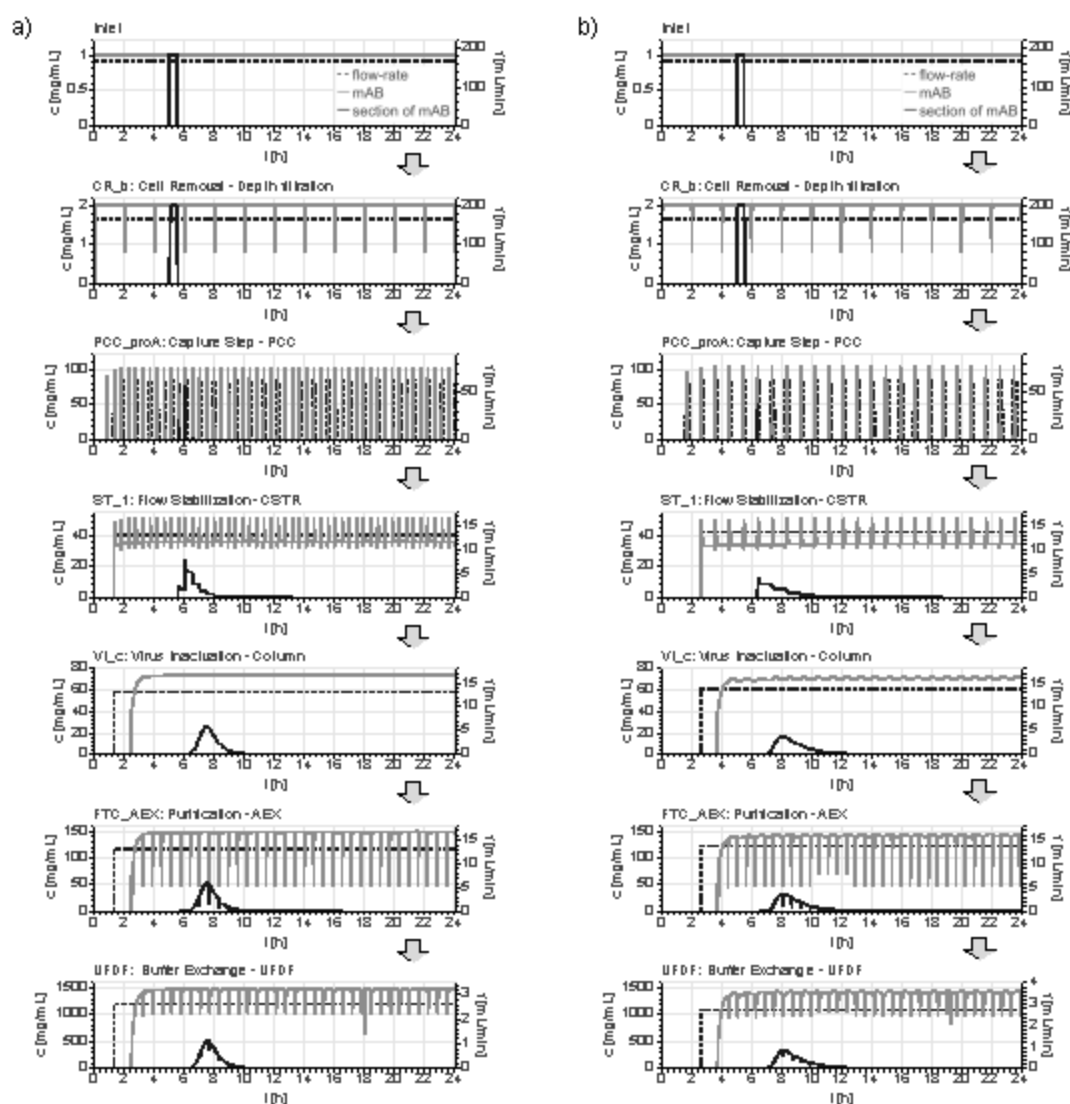
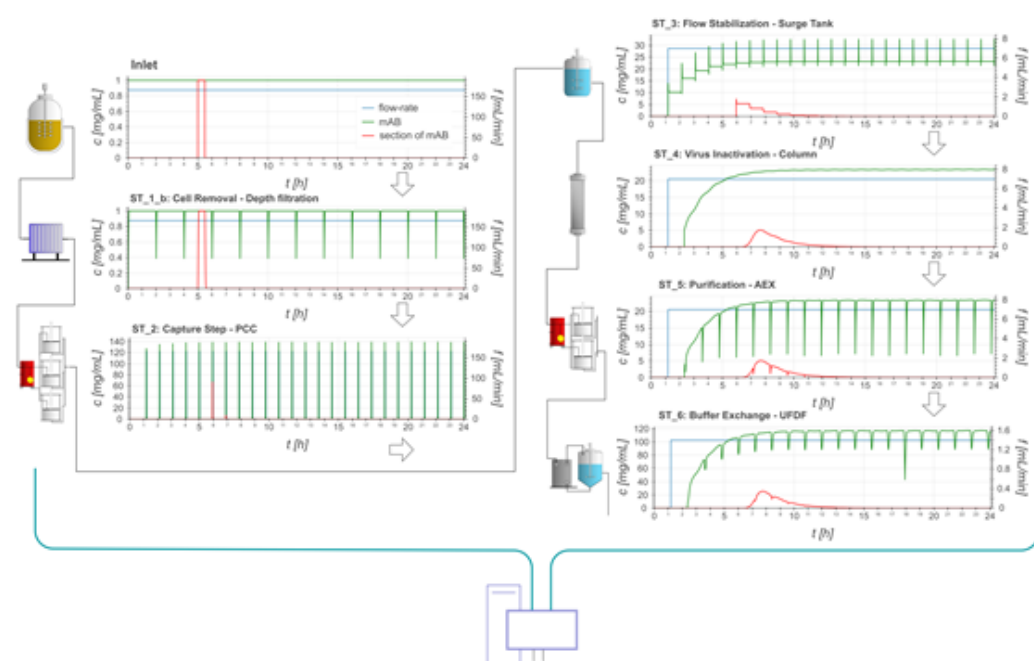


Figure 6. The effect of the PCC column size on the RTD of an integrated continuous process. A stream of product (gray line) with a labeled section (solid black line) are propagated throughout the integrated process **(A)**. Doubling the PCC column volume **(B)** results in double cycle duration at PCC step and a broader RTD downstream.

Graphical Abstract



A thorough understanding of mass flows in the process is required by the FDA in their draft on guidelines for continuous integrated biomanufacturing.

We have created a framework for modeling residence time distributions (RTD) for such processes.

The framework can produce fast and accurate RTD models based on available data about the process that is to be modeled and is publicly available under the open source license.